

```

1  #include <WiFi.h>
2  #include <WebServer.h>
3
4  // Wi-Fi credentials
5  const char* ssid = "IRONMAN1";
6  const char* password = "1234567890";
7
8  // Web server
9  WebServer server(80);
10
11 // LED pin
12 const int ledPin = 33;
13 bool isAlertActive = false;
14
15 void setup() {
16   // Initialize serial monitor
17   Serial.begin(115200);
18
19   // Initialize LED pin
20   pinMode(ledPin, OUTPUT);
21   digitalWrite(ledPin, LOW);
22
23   // Connect to Wi-Fi
24   WiFi.begin(ssid, password);
25   Serial.print("Connecting to Wi-Fi");
26   while (WiFi.status() != WL_CONNECTED) {
27     delay(500);
28     Serial.print(".");
29   }
30   Serial.println("\nConnected to Wi-Fi");
31   Serial.print("IP Address: ");
32   Serial.println(WiFi.localIP());
33
34   // Define server routes
35   server.on("/", handleMainPage);
36   server.on("/alerton", handleAlertPage);
37   server.on("/reset", handleResetPage);
38

```

Fig. 5: Snippet of the source code

calls should be the IndusBoard's IP address plus the hosted page.

For example:

`http://192.168.230.48/alerton`

IndusBoard Coin code implementation

The IndusBoard Coin acts as a web server that:

- Receives alerts from the Teachable Machine model.

- Controls an LED to indicate an engine fault.

- Hosts a simple HTML page for visual monitoring.

The code is created in Arduino IDE. First, configure the Wi-Fi SSID in the provided code by setting it with the network's Wi-Fi SSID and password. Then, upload the code to the board, obtain the IP address, and copy it. Replace the IP address in the HTML code for Teachable Machine, as shown earlier:

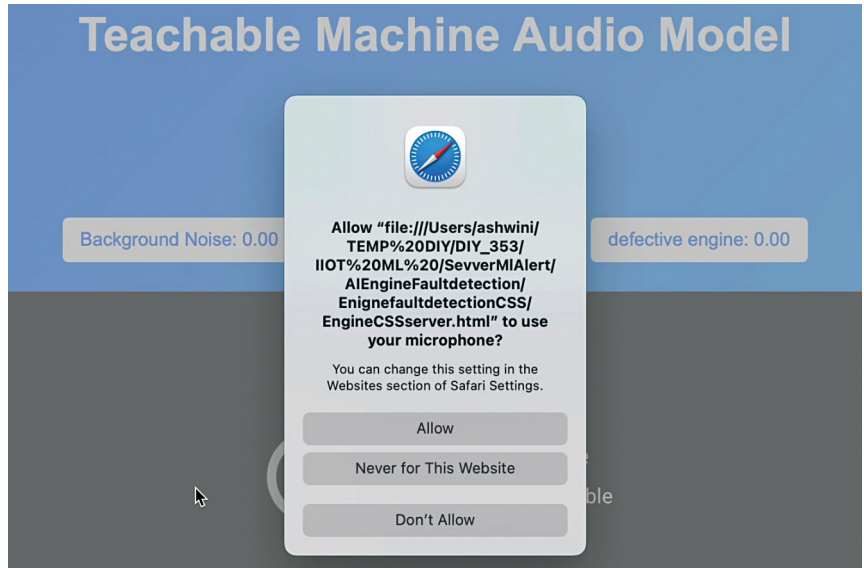


Fig. 6: ML asking mic permission and model predicting engine fault using sound

```

fetch('http://192.168.230.48/
alerton'); // Sends alert to Indus-
Board Coin.

```

Testing and running the system

Power the IndusBoard Coin using a 5V VIN pin or a USB port. Alternatively, connect it to a 3.3V pin. Wait for the coin to connect to the Wi-Fi network.

Run the Teachable Machine code on a laptop in a web browser.

Click the 'Start' button and grant microphone permission.

Once the microphone is granted access, the engine sound will be monitored.

If an abnormal sound is detected:

The IndusBoard Coin will send

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

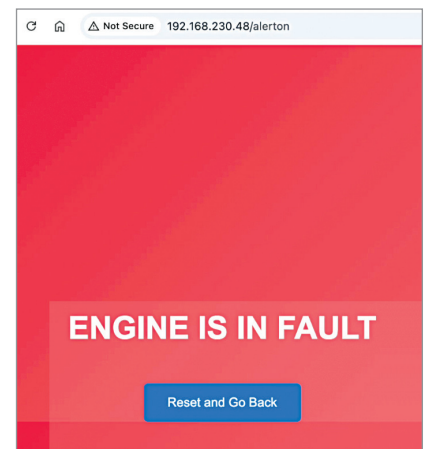


Fig. 7: Webpage output on engine fault and reset button

data to the laptop and display an alert.

The onboard alert LED on pin 33 will blink.

A buzzer can be added for additional alerts.

The webpage will indicate that the engine is faulty.

A new webpage with a 'Reset' button will appear on the laptop. Clicking the 'Reset' button will stop the alert. **EFY**

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